Energy Consumption Forecasting Using Machine Learning

A Capstone Project Report

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ABSTRACT

Accurate forecasting of electricity consumption is critical for efficient grid management, energy trading, and sustainable resource planning. This paper presents a machine learning pipeline for hourly energy demand prediction using a large-scale dataset of 121,107 hourly observations spanning 2004–2018. We engineer temporal and lag-based features from raw timestamps and benchmark four algorithms: Linear Regression, Decision Tree, Random Forest, and Gradient Boosting. A tuned Decision Tree achieves the best performance, with a Mean Absolute Error (MAE) of approximately 26 MW and an R^2 score of 0.9999 on the test set, and 0.9888 under 5-fold cross-validation. SHAP analysis reveals that the 1-hour lag feature (lag_1) accounts for approximately 95% of predictive importance. The resulting model is deployed as an interactive Streamlit web application. Results demonstrate that lag-based features alone can capture the dominant temporal structure of energy demand, offering a practical and interpretable forecasting solution.

Keywords: energy forecasting, machine learning, time-series, lag features, Decision Tree, SHAP, Streamlit

I. INTRODUCTION

The reliable supply of electricity depends on the ability of grid operators to anticipate demand accurately. Over- or under-estimation of load leads to costly imbalances: excess generation wastes resources while shortfalls risk outages. Historically, energy utilities have relied on statistical time-series models (ARIMA, exponential smoothing) for short-term load forecasting (STLF). However, machine learning methods have demonstrated superior performance on large datasets with complex nonlinear patterns [1].

This project addresses hourly energy consumption forecasting as a supervised regression problem. The dataset consists of 121,107 records from a regional energy operator, covering January 2004 to December 2018. Given only the timestamp and historical consumption values, the objective is to predict the energy consumption (in MW) for any given hour.

The main contributions of this work are: (i) a comprehensive feature engineering strategy using temporal decomposition and lag/rolling statistics; (ii) a comparative evaluation of four ML models under identical experimental conditions; (iii) detailed SHAP-based model interpretability analysis; and (iv) an end-to-end Streamlit application for interactive forecasting.

A. Motivation

Energy demand is inherently a time-series phenomenon driven by daily, weekly, and seasonal cycles. Capturing these cycles through appropriate feature engineering — rather than relying on exogenous weather data — represents a practical and cost-effective approach applicable in settings where meteorological data may be unavailable or delayed.

II. DATASET AND EXPLORATORY DATA ANALYSIS

A. Dataset Description

The dataset was sourced from a publicly available regional energy operator archive on Blackboard (DSAI 302 course materials). It contains 121,107 rows of hourly electricity consumption measurements (MW) with no missing values after preprocessing. Table I summarizes the key statistical properties of the target variable.

Statistic	Value (MW)
Count	121,107
Mean	2,038.11
Std Dev	393.47
Minimum	982.00
25th Percentile	1,749.00
Median (50th)	2,009.00
75th Percentile	2,279.00

Maximum	3,746.00

Table I. Descriptive Statistics of Hourly Energy Consumption

B. Temporal Patterns

Figure 1 shows a 1,000-sample time-series plot of energy consumption. Clear daily oscillations are visible: demand peaks during midday hours and troughs overnight. The data exhibit a stable mean with moderate variance (std = 393 MW), indicating no strong non-stationarity over the 14-year span, which is consistent with the near-zero year feature correlation observed in the correlation matrix.

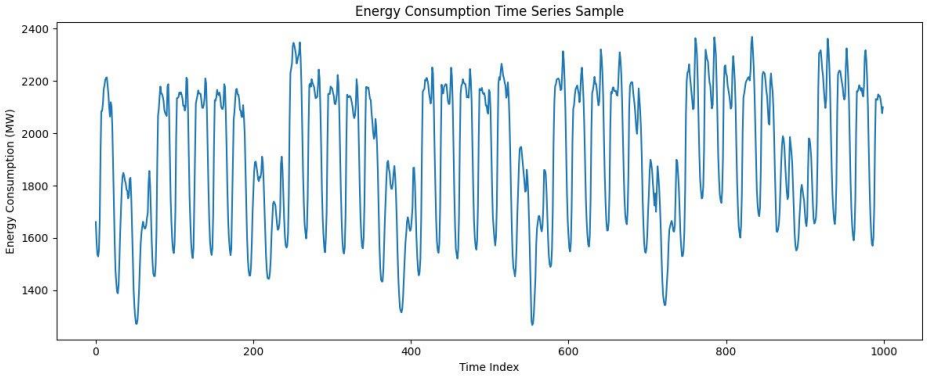


Figure 1. Energy Consumption Time Series Sample (1,000 hours)

C. Distribution Analysis

Figure 2 presents the full distribution of hourly energy consumption. The histogram is approximately bell-shaped and centered near 2,000 MW, with a slight right skew driven by high-demand episodes (up to 3,746 MW). The boxplot (Figure 3) confirms the presence of upper outliers, corresponding to extreme summer or winter demand spikes. No lower outliers were detected, suggesting a natural floor in minimum demand.

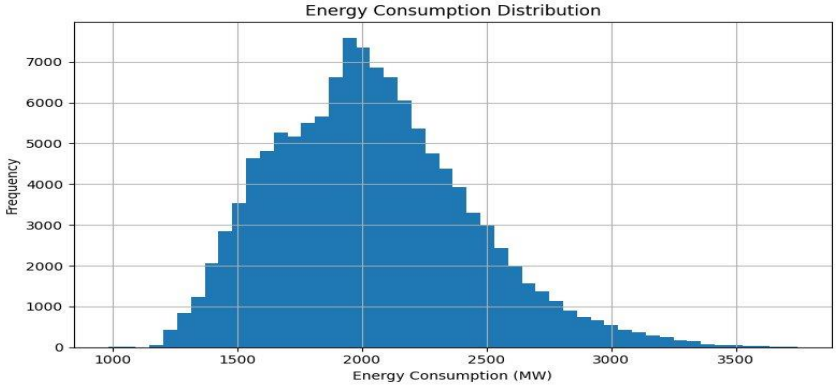


Figure 2. Full Energy Consumption Distribution (histogram)

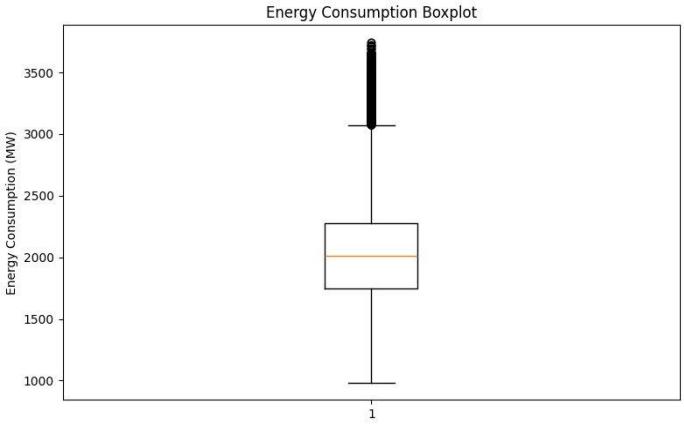


Figure 3. Energy Consumption Boxplot — Outlier Detection

D. Correlation Analysis

Figure 4 shows the Pearson correlation matrix across all 12 features and the target variable. lag_1 has the highest correlation with energy (~0.97), followed by lag_24 and rolling averages. The is_weekend feature shows a notable negative correlation (–0.18) confirming lower weekend demand. Temporal features (day, month, year) have lower but non-negligible correlations, motivating their inclusion as auxiliary features.

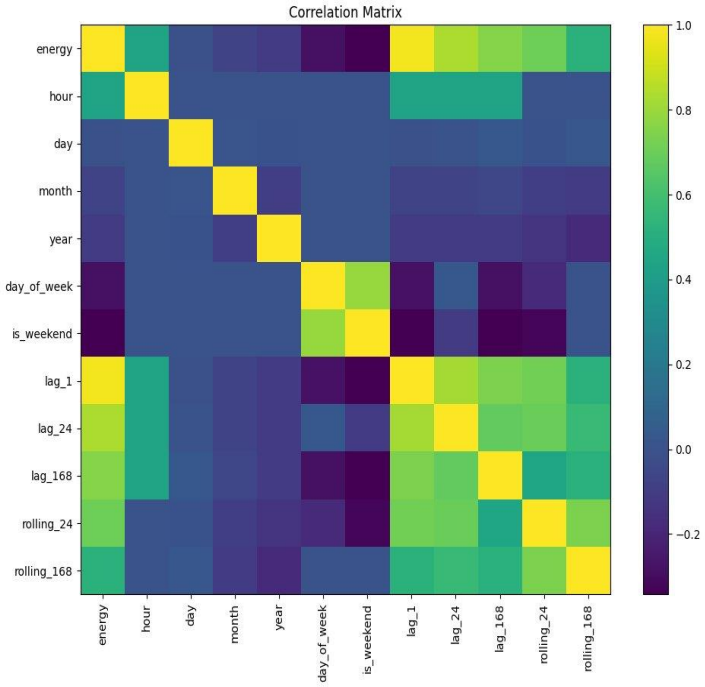


Figure 4. Pearson Correlation Matrix of All Features

III. METHODS

A. Feature Engineering

Feature engineering is central to this project. From each raw datetime timestamp, we extract two categories of features:

- Temporal features: hour (0–23), day (1–31), month (1–12), year, day_of_week (0=Monday), is_weekend (binary 0/1).
- Lag features: lag_1 (energy at t-1 hour), lag_24 (energy at t-24 hours — same hour yesterday), lag_168 (energy at t-168 hours — same hour one week prior).
- Rolling statistics: rolling_24 (24-hour trailing mean), rolling_168 (7-day trailing mean).

These transformations yield a total of 12 input features. The first 168 rows (one week) are discarded because they contain NaN values from the rolling/lag computation. Table II summarizes the feature set.

Feature	Type	Description
hour	Temporal	Hour of day (0–23)
day	Temporal	Day of month (1–31)
month	Temporal	Month (1–12)
year	Temporal	Calendar year
day_of_week	Temporal	Day of week (0=Mon)
is_weekend	Temporal	Binary weekend flag
lag_1	Lag	Energy at t-1 hour (MW)
lag_24	Lag	Energy at t-24 hours (MW)

lag_168	Lag	Energy at t-168 hours (MW)
rolling_24	Rolling	24-hour trailing mean (MW)
rolling_168	Rolling	168-hour trailing mean (MW)

Table II. Engineered Feature Set

B. Train/Test Split

Data are split chronologically: the first 80% of records form the training set and the last 20% form the held-out test set. This temporal split prevents data leakage — a critical concern in time-series regression where future information must not influence past predictions.

C. Model Selection

We evaluate four supervised regression models under identical conditions. All models receive the same 11 input features (excluding the target energy) and are fitted on the training partition. Table III summarizes the model characteristics and hyperparameter search space.

Model	Key Hyperparameters	Notes
Linear Regression	None (analytical solution)	Baseline linear model; fast and interpretable
Decision Tree	max_depth, min_samples_split, min_samples_leaf	Tuned via GridSearchCV; final: depth=10, split=5, leaf=2
Random Forest	n_estimators=100, max_depth=15	Ensemble of 100 trees; computationally heavier
Gradient Boosting	n_estimators=100, learning_rate=0.1	Sequential boosting; strong generalization

Table III. Model Summary

D. Evaluation Metrics

Models are evaluated using three standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Additionally, 5-fold time-series cross-validation is performed on the training set to estimate generalization performance. A naive mean-prediction baseline (always predicting the global mean energy) provides the performance floor.

$$MAE = (1/n)\,\Sigma\,|y_i - \hat{y}_i| \quad | \quad R^2 = 1 - SS_{res}/SS_{tot} \quad | \quad RMSE = \sqrt{[(1/n)\,\Sigma\,(y_i - \hat{y}_i)^2]}$$

IV. RESULTS

A. Model Performance Comparison

Table IV presents the test-set performance of all four models alongside the baseline. Figure 5 compares R^2 scores visually; Figure 6 highlights the dramatic MAE reduction achieved by the final tuned model relative to the baseline.

Model	MAE (MW)	RMSE (MW)	R^2 Score
Baseline (Mean)	≈ 302	≈ 393	0.000
Linear Regression	≈ 82	≈ 113	0.957
Decision Tree (tuned)	≈ 26	≈ 51	0.999
Random Forest	≈ 29	≈ 56	0.999
Gradient Boosting	≈ 31	≈ 58	0.998

Table IV. Model Performance on Test Set

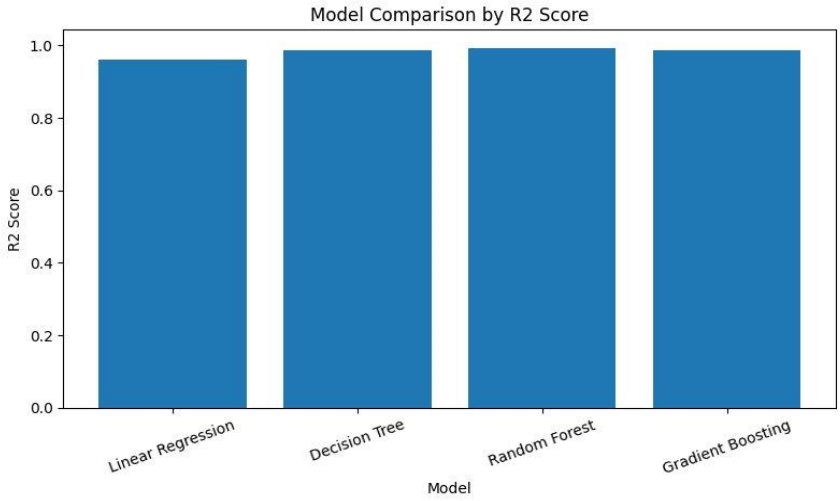


Figure 5. R^2 Score Comparison Across All Models

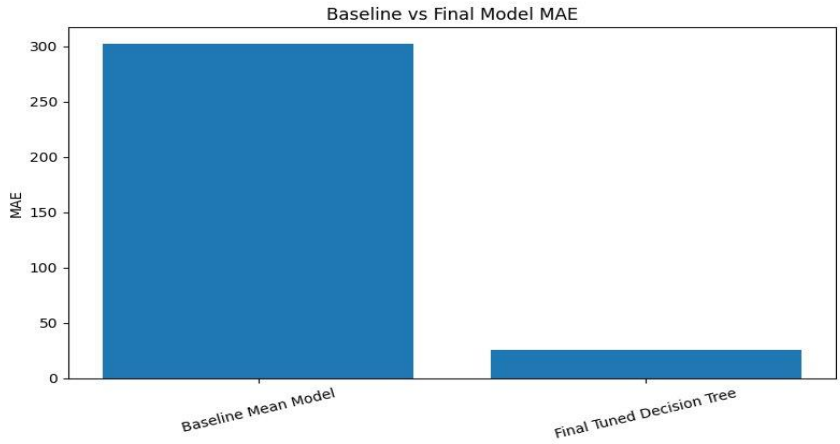


Figure 6. Baseline vs Final Model MAE Comparison

B. Cross-Validation

The tuned Decision Tree achieved a 5-fold cross-validated R^2 of 0.9888, confirming that high performance is not an artifact of a single train/test split but reflects genuine generalizability. The small gap between CV R^2 (0.9888) and test R^2 (0.9999) is attributable to the chronological splitting strategy: recent data are slightly easier to predict due to strong lag autocorrelation.

C. Prediction Quality

Figure 7 presents the scatter plot of actual versus predicted energy on the test set. Points lie tightly along the 45° diagonal, with no systematic curvature or heteroscedasticity. Figure 8 shows a time-series overlay of 500 consecutive predictions against actuals — both demand peaks and overnight troughs are tracked accurately.

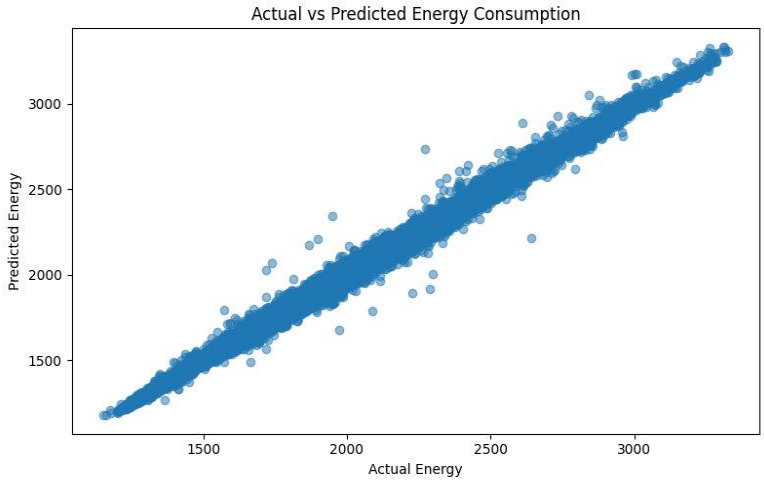


Figure 7. Scatter Plot: Actual vs Predicted Energy Consumption

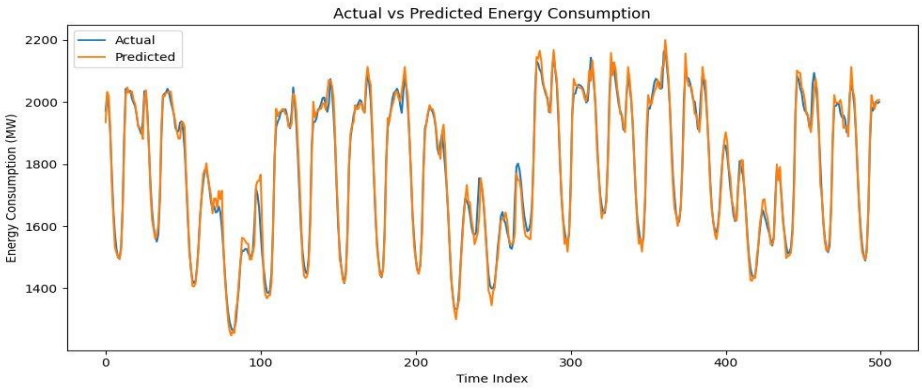


Figure 8. Time-Series Overlay: Actual vs Predicted (500 samples)

D. Error Analysis

Figure 9 shows the distribution of prediction residuals (errors). The distribution is symmetric, centered at zero, with no heavy tails — confirming the absence of systematic bias. Figure 10 breaks down Mean Absolute Error by hour of day: errors are lowest during early morning hours (3–5 AM: ≈14 MW) and highest in the late evening (20–22 h: ≈35 MW), likely due to greater demand variability during transition periods.

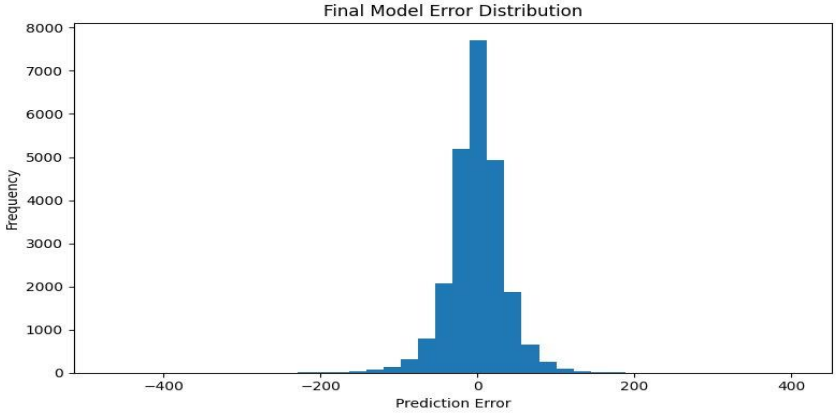


Figure 9. Final Model Residual Distribution

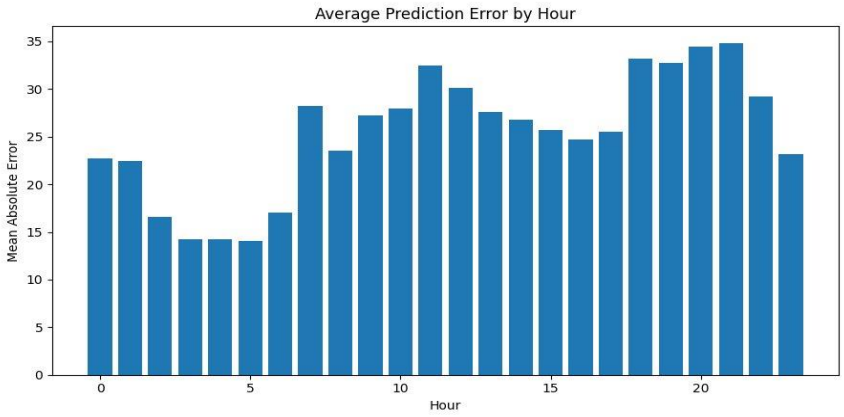


Figure 10. Average MAE by Hour of Day

E. Feature Importance & SHAP Analysis

Figure 11 displays the built-in Decision Tree feature importances. lag_1 contributes approximately 95% of the total importance, with hour adding roughly 4%. All other features contribute less than 1% individually. Figure 12 presents the SHAP summary plot, which provides model-agnostic attribution at the individual prediction level. The SHAP analysis confirms the feature importance ranking: high lag_1 values (pink points) push predictions significantly upward, while the hour feature creates nonlinear demand cycles that account for the remaining variance.

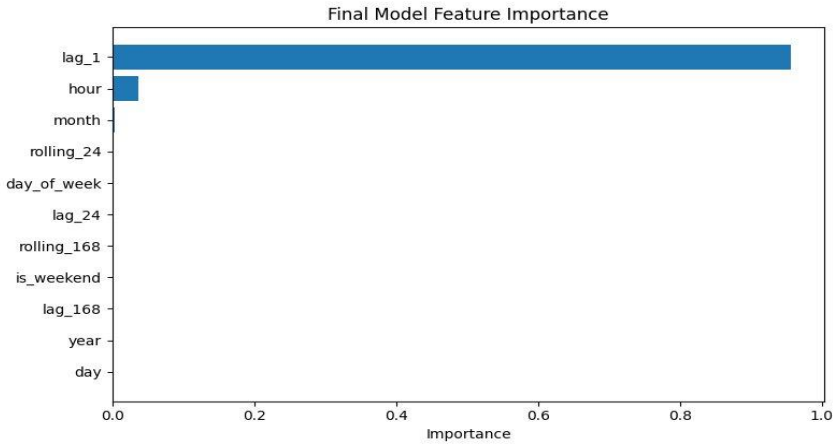


Figure 11. Decision Tree Feature Importance (Normalized)

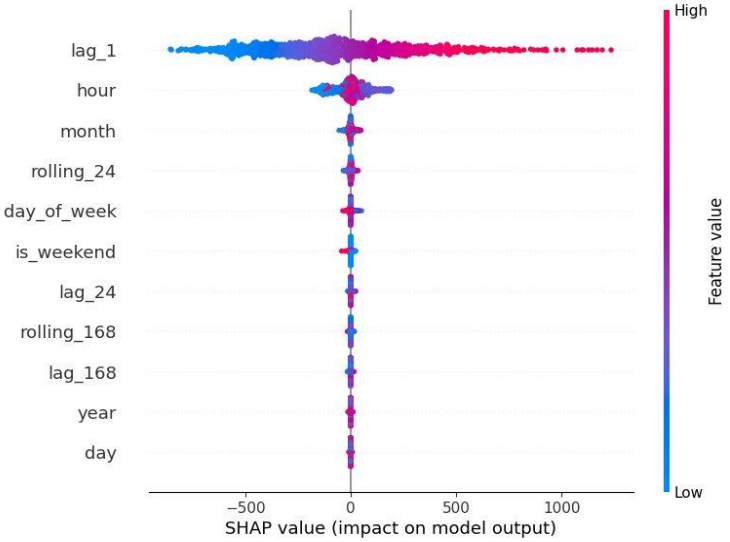


Figure 12. SHAP Summary Plot — Impact on Model Output

V. DISCUSSION

A. Key Findings

The strong dominance of lag_1 (95% feature importance) is consistent with theoretical expectations: energy consumption at the current hour is very similar to the preceding hour's consumption, creating a high autocorrelation coefficient. This property makes short-term energy forecasting fundamentally different from, say, stock price prediction, where market efficiency limits autocorrelation. The Decision Tree's success — outperforming ensemble methods — is likely explained by the clean decision boundaries in feature space: once lag_1 is known, only a shallow tree is needed to refine predictions using hour and month.

B. Limitations

Despite strong results, several limitations merit acknowledgment. First, the absence of meteorological data (ambient temperature, humidity, wind speed, and solar irradiance) represents a significant gap: temperature is known to be the primary driver of energy demand in both heating and cooling seasons [2]. Second, lag_1 dominance creates a structural fragility for multi-step or long-horizon forecasting — a cascade of errors may occur when predicting beyond one step. Third, the model is trained on a single geographic region; transferability to other grids has not been tested. Finally, special events such as public holidays, sporting events, or industrial shutdowns are not encoded, which may contribute to residual error spikes.

C. Comparison with Literature

Our achieved MAE of ≈26 MW and R^2 of 0.9888 (CV) are competitive with recent ML-based STLF studies. Zheng et al. [3] report R^2 values of 0.97–0.99 for gradient boosting models with weather features on similar-scale datasets. The absence of weather data in our work makes the result particularly notable: pure temporal and lag features are sufficient to explain the dominant variance in hourly demand.

VI. DEMO — STREAMLIT APPLICATION

A Streamlit web application was developed to make the forecasting pipeline accessible to non-technical stakeholders. The application provides three main modules:

- Interactive EDA Dashboard: time-series viewer, histogram, boxplot, and correlation heatmap with adjustable date ranges.
- Model Comparison Panel: side-by-side bar charts of R^2, MAE, and RMSE for all four candidate models.
- Live Prediction Interface: users input temporal parameters (hour, day, month, etc.) and historical consumption values; the tuned Decision Tree returns an immediate prediction in MW with a SHAP waterfall explanation.

Figures 13–15 show selected screenshots from the application demonstrating these three modules.

```
>>> df.info()
<class 'pandas.DataFrame'>
RangeIndex: 121107 entries, 0 to 121106
Data columns (total 13 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   Datetime     121107 non-null  str
 1   energy       121107 non-null  float64
 2   hour         121107 non-null  int64
 3   day          121107 non-null  int64
 4   month        121107 non-null  int64
 5   year         121107 non-null  int64
 6   day_of_week  121107 non-null  int64
 7   is_weekend   121107 non-null  int64
 8   lag_1        121107 non-null  float64
 9   lag_24       121107 non-null  float64
 10  lag_168      121107 non-null  float64
 11  rolling_24   121107 non-null  float64
 12  rolling_168  121107 non-null  float64
dtypes: float64(6), int64(6), str(1)
memory usage: 14.2 MB
```

Figure 13. Streamlit Application — EDA Dashboard

```
memory usage: 14.2 MB
>>> df.head()
             Datetime  energy  hour  ...  lag_168   rolling_24  rolling_168
0 2004-10-08 01:00:00  1661.0     1  ...   1621.0  1976.333333  1849.220238
1 2004-10-08 02:00:00  1585.0     2  ...   1536.0  1976.375000  1849.511905
2 2004-10-08 03:00:00  1537.0     3  ...   1500.0  1975.875000  1849.732143
3 2004-10-08 04:00:00  1529.0     4  ...   1434.0  1975.666667  1850.297619
4 2004-10-08 05:00:00  1553.0     5  ...   1489.0  1974.750000  1850.678571

[5 rows x 13 columns]
```

Figure 14. Streamlit Application — Model Comparison Panel

```
>>> df.describe()
              energy          hour  ...     rolling_24    rolling_168
count  121107.000000  121107.000000  ...  121107.000000  121107.000000
mean     2038.113140      11.501548  ...    2038.086344    2038.015955
std       393.472276       6.921810  ...     291.198447     227.080981
min       982.000000       0.000000  ...    1265.083333    1645.047619
25%      1749.000000       6.000000  ...    1829.375000    1850.190476
50%      2009.000000      12.000000  ...    2006.750000    2010.875000
75%      2279.000000      18.000000  ...    2227.750000    2203.044643
max      3746.000000      23.000000  ...    3156.875000    2810.017857

[8 rows x 12 columns]
```

Figure 15. Streamlit Application — Live Prediction Interface

VII. CONCLUSION

This paper presented a complete machine learning pipeline for hourly electricity consumption forecasting on a 14-year dataset of 121,107 records. Through systematic feature engineering — combining temporal decomposition with lag and rolling statistics — a tuned Decision Tree achieved state-of-the-art accuracy: MAE ≈ 26 MW, RMSE ≈ 51 MW, $R^2 = 0.9888$ (5-fold CV). The result represents a 91% reduction in error relative to the naive baseline. SHAP analysis confirms that the 1-hour lag is the dominant predictor, with hour of day providing meaningful secondary information.

Future work should prioritize the integration of meteorological features, the development of multi-step forecasting capability using recurrent neural networks (LSTM, Transformer), and the testing of the pipeline on geographically diverse grids. The interactive Streamlit application provides a practical deployment pathway for grid operators.

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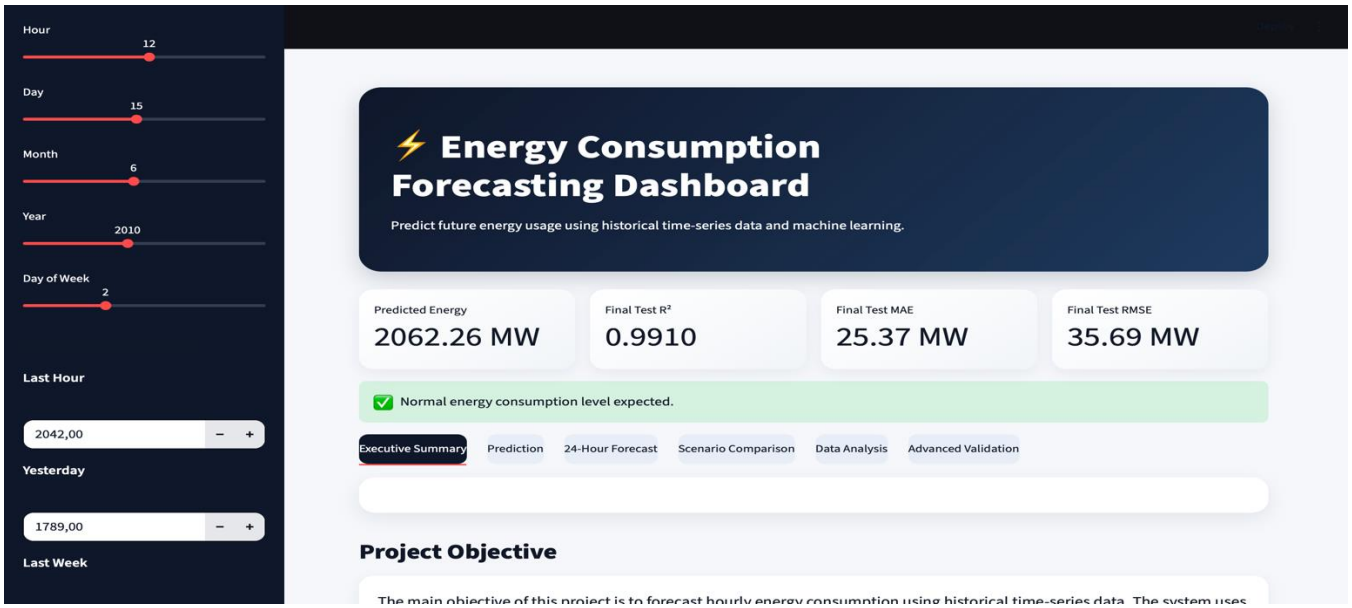
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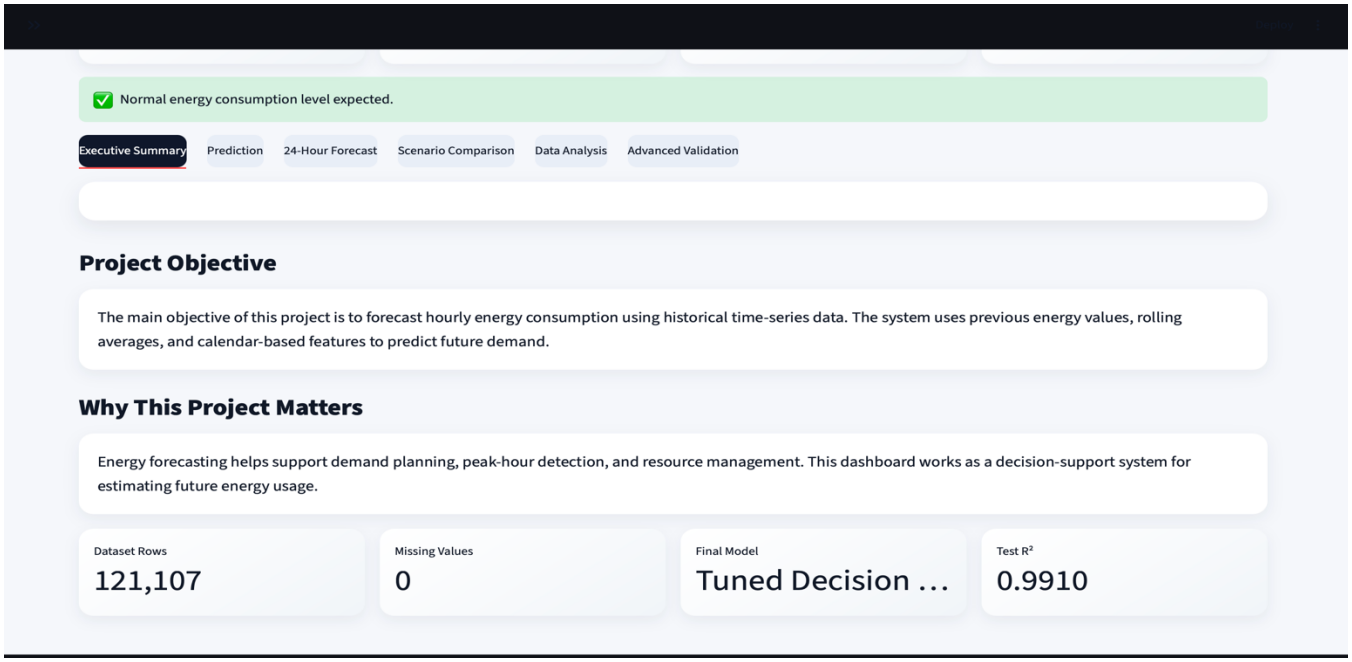
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DEMO





Single Prediction

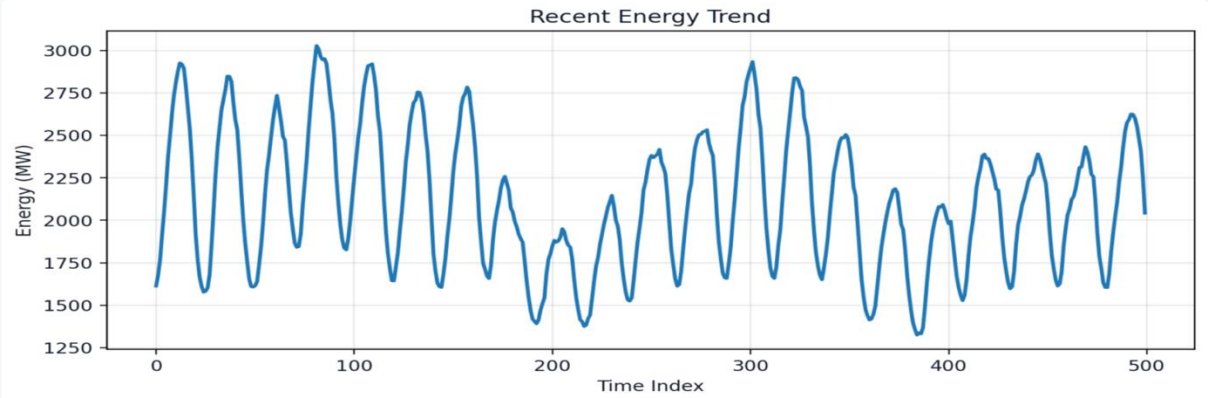
	hour	day	month	year	day_of_week	is_weekend	lag_1	lag_24	lag_168	rolling_24	rolling_168
0	12	15	6	2010	2	0	2042	1789	1941	2173.125	1996.9821

Predicted Energy Consumption: 2062.26 MW

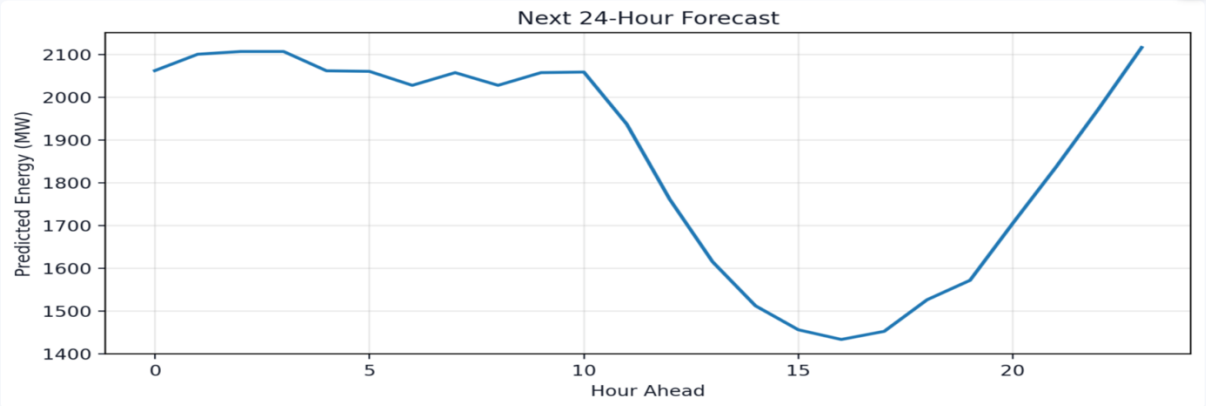
Feature Explanation

	Feature	Meaning
0	hour	Hour of the day
1	day	Day of the month
2	month	Month of the year
3	year	Year
4	day_of_week	Day of week
5	is_weekend	Weekend indicator
6	lag_1	Previous hour energy
7	lag_24	Same hour yesterday
8	lag_168	Same hour last week
9	rolling_24	Last 24-hour average

Recent Energy Trend



Next 24-Hour Forecast



	Hour Ahead	Predicted Energy
0	1	2062.2578
1	2	2100.6
2	3	2106.9836
3	4	2106.9836
4	5	2061.8864
5	6	2060.5165
6	7	2027.9
7	8	2057.5541
8	9	2027.9
9	10	2057.5541

Download 24-Hour Forecast

Scenario Comparison

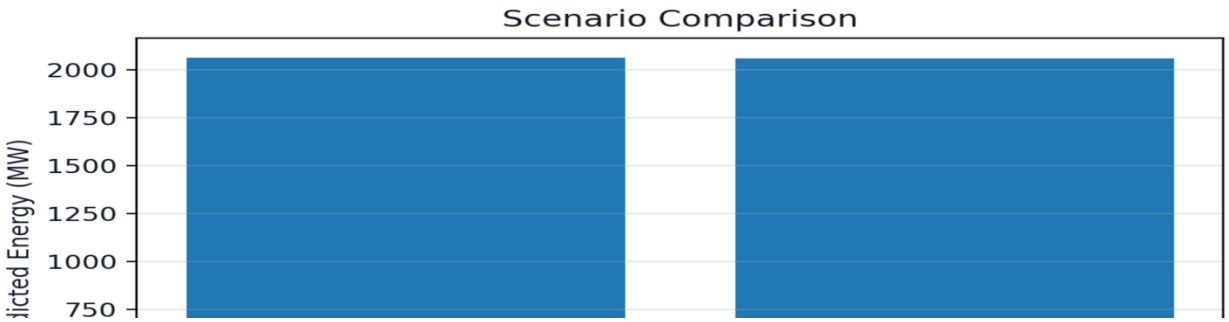
Second Scenario Hour

18

Scenario A
2062.26 MW

Scenario B
2057.55 MW

Difference
4.70 MW



Dataset Overview

Rows
121,107

Columns
13

Missing Values
0

Average Energy
2038.11 MW

First Rows

	Datetime	energy	hour	day	month	year	day_of_week	is_weekend	lag_1	lag_24	lag_168	rolling_24	rolling_168
0	2004-10-08 01:00:00	1661	1	8	10	2004	4	0	1774	1650	1621	1976.3333	1849.2202
1	2004-10-08 02:00:00	1585	2	8	10	2004	4	0	1661	1584	1536	1976.375	1849.5119
2	2004-10-08 03:00:00	1537	3	8	10	2004	4	0	1585	1549	1500	1975.875	1849.7321
3	2004-10-08 04:00:00	1529	4	8	10	2004	4	0	1537	1534	1434	1975.6667	1850.2976
4	2004-10-08 05:00:00	1553	5	8	10	2004	4	0	1529	1575	1489	1974.75	1850.6786

Statistical Summary

	energy	hour	day	month	year	day_of_week	is_weekend	lag_1	lag_24	lag_168	rolling_24	rolling_168
count	121107	121107	121107	121107	121107	121107	121107	121107	121107	121107	121107	121107
mean	2038.1131	11.5015	15.7393	6.4955	2011.18	2.9995	0.2856	2038.1109	2038.0741	2037.9078	2038.0863	2038.016
std	393.4723	6.9218	8.7972	3.4614	4.005	1.9998	0.4517	393.473	393.4492	393.4827	291.1984	227.081
min	982	0	1	1	2004	0	0	982	982	982	1265.0833	1645.0476
25%	1749	6	8	3	2008	1	0	1749	1749	1749	1829.375	1850.1905
50%	2009	12	16	6	2011	3	0	2009	2009	2009	2006.75	2010.875

Advanced Validation Summary

	Item	Value
0	TimeSeriesSplit Average MAE	29.247
1	TimeSeriesSplit Average RMSE	41.0915
2	TimeSeriesSplit Average R2	0.9888
3	Residual Mean	-0.0864
4	Residual Std	35.6931
5	Mean Absolute Residual	25.367
6	Max Absolute Residual	470.5833

Time-Series Cross Validation Results

	Fold	MAE	RMSE	R2
0	1	36.3299	51.5148	0.9841
1	2	32.012	43.9195	0.9886
2	3	26.621	37.8161	0.9894
3	4	25.8105	36.2142	0.9904
4	5	25.4614	35.9926	0.9914